Artificial Intelligence
UNSW Sydney
Topic exercises: Artificial neural networks

Question 1: McCulloch-Pitts' model

Consider the logic function AND with 4 inputs. Implement a McCulloch-Pitts' model that produces this result in its output. Assume $w_i = 1$.

Question 2: Forward propagation

Consider a single-layer perceptron with 2 inputs and 4 neurons with an activation function sgn or threshold logic function. Moreover, take into account inputs, weight matrix, and thresholds shown next. What would be the network output for these inputs?

$$W_{ij} = \begin{bmatrix} 0.5 & 1.0 & 0.6 \\ -0.5 & -0.7 & 0.1 \\ 0.25 & 0.1 & 0.0 \\ 0.55 & -1.1 & -0.2 \end{bmatrix} \quad X = \begin{bmatrix} 12 & -6 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.5 & 0 & 1.2 & 0.2 \end{bmatrix}$$

Question 3: Perceptron learning

Consider the training data shown in Table 1 to divide the space as shown in Fig. 1. Demonstrate the single-layer perceptron learning rule using a learning rate $\alpha = 1.0$ and initial weights $w_1 = 0$, $w_2 = -1$, and $b = -2.5$.

Table 1: Training examples and classes.

Training example	x_1	x_2	Class
a	0.0	1.0	-1
b	2.0	0.0	-1
c	1.0	1.0	1

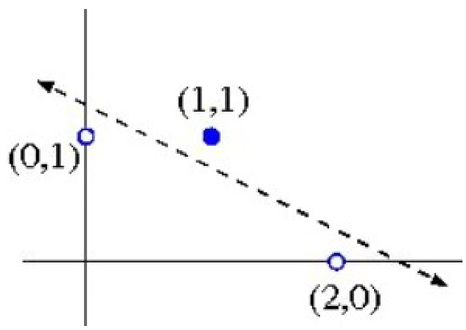

Figure 1: Space division of the training examples.

Question 4: Neural desing

- What would be the MLP architecture to approximate a non-linear function of 3 inputs and 2 outputs if you have available 3,000 samples? Consider 70% data for training and 30% for validation.

- What would be the MLP architecture to approximate a non-linear function of 6 inputs and 3 outputs if you have available 100 samples? Consider 80% data for training and 20% for validation.

Question 5: Computing any Logical Function with a 2-layer Network

Recall that any logical function can be converted to Conjunctive Normal Form (CNF), which means a conjunction of terms where each term is a disjunction of (possibly negated) literals. This is an example of an expression in CNF:

$$(A \vee B) \wedge (\neg B \vee C \vee \neg D) \wedge (D \vee \neg E)$$

Assuming False=0 and True=1, explain how each of the following could be constructed. You should include the bias for each node, as well as the values of all the weights (input-to-output or input-to-hidden and hidden-to-output, as appropriate).

- Perceptron to compute the OR function of m inputs,
- Perceptron to compute the AND function of n inputs,
- Two-layer Neural Network to compute the function

$$(A \vee B) \wedge (\neg B \vee C \vee \neg D) \wedge (D \vee \neg E)$$

With reference to this example, explain how a two-layer neural network could be constructed to compute any (given) logical expression, assuming it is written in Conjunctive Normal Form.
Hint: first consider how to construct a Perceptron to compute the OR function of m inputs, with k of the m inputs negated.

Question 6: Backpropagation

Consider the neural network architecture shown in Fig. 2 for a binary classification problem. The values for weights and biases are shown in the figure. We define:

$a_1 = w_{11}x_1 + b_{11}$
$a_2 = w_{12}x_1 + b_{12}$
$a_3 = w_{21}z_1 + w_{22}z_2 + b_{21}$
$z_1 = ReLU(a_1), ReLU(x) = max(0, x)$
$z_2 = ReLU(a_2)$
$z_3 = \sigma(a_3), \sigma(x) = \frac{1}{1+e^{-x}}$
$Loss = \frac{1}{2}(y - z_3)^2$

Calculate the new bias b_{11} after one iteration, after performing backpropagation on the bias for a data sample ($x_1 = 1, y = 1$) and learning rate equals to 1.

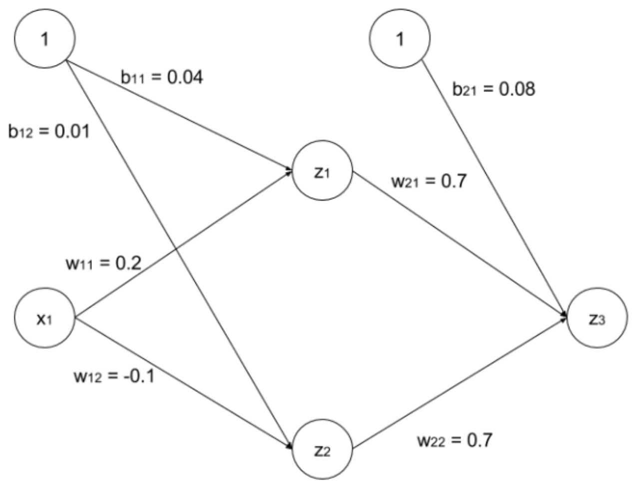

Figure 2: Neural network architecture.